

Neal Daftary

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EDUCATION

Institute of Technology, Nirma University

Ahmedabad, India

B.Tech in Computer Science and Engineering (AI & ML) - CGPA 7.77

Sep 2024 – Jun 2028 (Expected)

- **Student Chairperson - ACM NU** (Sept 2025 - Present): Grew the student chapter to **200+ members** and launched 2 flagship programs, including a prompt engineering hackathon and a 4-track mentorship pipeline.

WORK EXPERIENCE

Student Researcher - ISRO Funded Project

May 2026 – Present

Institute of Technology, Nirma University

Ahmedabad, India

- Developing a specialized Computer Vision system for the morphological analysis and mapping of lunar surface features.

Software Engineering Intern

Oct 2025 – Dec 2025

MZHubtech

Hamburg, Germany (Remote)

- Achieved **+57% performance** and **+40% SEO** gains (Core Web Vitals) by deploying an SSR **Next.js** app to **Azure App Service** via a GitHub Actions CI/CD pipeline.
- Enabled scalable data persistence and automated transactional emails for production traffic by integrating **Azure Cosmos DB** (NoSQL) and **SendGrid** SMTP.
- Delivered **sub-600ms AI response times** across 16,600+ embedded vectors by engineering a hybrid FAISS and BM25 RAG pipeline powered by Llama 3.3 70B via Groq.
- Maintained strict output reliability and mitigated hallucination risks by engineering multi-tiered safety filters and a comprehensive **LangSmith** observability stack.

AI Intern

Jun 2025 – Sep 2025

8xSports

Bhilwara, India (Remote)

- Engineered a FAISS IVF-PQ/OPQ index achieving **sub-800ms retrieval latency for 15,000 profiles** through GPU batch encoding.
- **Reduced edge-device inference to sub-20ms** by fine-tuning YOLOv8s with custom augmentations and optimizing the export pipeline with **ONNX** and **TensorRT FP16**.
- Enhanced athlete re-identification robustness under occlusion by fusing **DINOv2** patch embeddings and LBP texture histograms.

OPEN SOURCE CONTRIBUTIONS

Google DeepMind ([jax_privacy](#))

Jul 2026 – Present

Open Source Contributor

- Refactored the chex dependency to a test-only scope, trimming the core install footprint without changing runtime behavior.
- Improved core library ergonomics and documentation clarity.
- Added pyrely static type checking alongside the existing pytype checks in CI.

PROJECTS

HelioOps: Real-Time Space-Weather Advisory Platform | [LightGBM](#), [AgentScope](#), [Next.js](#) | [GitHub](#)

2026

- Predicted GPS L1 positioning error and HF-blackout probability with **95%+ CI coverage** (R2 0.986/0.958) by training **6 LightGBM** quantile-regression models on solar-wind and flare telemetry.
- Delivered regulatory-compliant advisories to **4 industries in under 60s** by architecting a 4-layer real-time pipeline fusing coronagraph imagery, NASA DONKI, GOES X-ray, and DSCOVR L1 data.

NavkaarAI: AI WhatsApp Agent Platform for Shopify | [Next.js](#), [BullMQ](#), [Redis](#), [Supabase](#) | [GitHub](#)

2026

- Recovered abandoned carts and automated support at **20K+ messages/mo per merchant** by integrating Meta's WhatsApp Cloud API with Anthropic Claude over Shopify OAuth.
- Sustained **>90% message delivery** on a horizontally scalable async backend by routing sends through **BullMQ** workers on Railway with Upstash Redis and Supabase Postgres.
- Resolved 10+ pre-launch issues via CI lint/typecheck and Sentry monitoring in a production **Turborepo** monorepo.

mcptail: Zero-Config Observability Tool for MCP Servers | [TypeScript](#), [Preact](#), [Vite](#) | [GitHub](#)

2026

- Replaced manual console debugging with a single-command wiretap across Claude Code, Cursor, and VS Code, capturing **100% local telemetry** with no external cloud dependencies.
- Exposed per-tool token cost and **p50/p95 latency** in real-time by building a live SSE dashboard with searchable timelines and error filtering.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL

Backend: FastAPI, Next.js, Node.js, Docker, Redis (BullMQ), PostgreSQL, REST/SSE

ML/CV: PyTorch, ONNX, TensorRT, FAISS, YOLOv8, SegFormer, DINOv2, SAM2, OpenCV, HuggingFace

LLM/MLOps: LangChain, LangGraph, LangSmith, MLflow, DVC, Airflow, GitHub Actions

Cloud/Tools: Azure, AWS (S3, ECR, ECS, SageMaker), Vercel, Railway, Claude Code

CERTIFICATES & ACHIEVEMENTS

CatBoost-Driven Anomaly Detection ([IEEE Xplore](#)): Published in *IEEE Sensor Letters*. Achieved **97.20% precision** on 200K+ samples, outperforming baselines.

4th National Rank, Mitsubishi Electric Cup 2026: Improved defect segmentation accuracy to **92.35%** by developing a DINOv2 + FPN-UNet system. [GitHub](#)

Track Winner & Top-5 National Finalist, HACKaMINeD 2026: Built LUMIN.AI, an explainable ML system for predictive maintenance out of 2,200+ participants. [GitHub](#)

Complete MLOps Bootcamp(Udemy) (certificate) & **Advanced Learning Algorithms**, DeepLearning.AI/Stanford (certificate): **99.77%**.